

NEXUS ■

NESCOE HACKATHON 2026 • Replit Development Prompt

THE CITY, CONNECTED.

Build NEXUS, a futuristic Smart City Dashboard for our college hackathon. NEXUS is a unified frontend interface that brings important city systems together into one intelligent dashboard.

Core systems: Traffic • Mobility • Energy • Environment • Waste • Weather • City Network • System Status

■ Technical Rules

Frontend only. Use HTML, CSS, JavaScript and React. Do not add sign in, sign up, logout, authentication, Firebase, databases, heavy backend, Node/Express API servers, payment systems, admin panels, user accounts, or complex server infrastructure.

The project must remain a lightweight, demonstrable college hackathon prototype. Do not introduce unnecessary technologies simply because they exist.

■ Do Not Make It Look Hardcoded

Do not scatter values throughout JSX. Keep demonstration data in clean centralized frontend data structures and render the interface from that data. Use reusable React components and frontend state.

```
const cityData = {
  traffic: { flow: 72, status: "Moderate" },
  weather: { temperature: 29, humidity: 68, wind: 14 },
  energy: { consumption: 438, renewable: 42 },
  environment: { aqi: 42, greenCover: 38 },
  mobility: { metro: 94, bus: 87, bike: 62 },
  waste: { efficiency: 76, recycling: 54 }
};
```

■ Design Direction

Make NEXUS feel like a real futuristic city operating system, not a generic college dashboard.

Visual style: near-black background, white typography, one neon accent, subtle grey grid, thin borders, dark glass panels, soft glow, huge typography, premium spacing, smooth animations and data-heavy UI.

Avoid: purple-heavy design, excessive gradients, cartoonish UI, generic SaaS appearance, huge rounded cards everywhere, excessive shadows and clutter.

■ HOME / HERO

NEXUS // SMART CITY

THE CITY,
CONNECTED.

A unified interface for understanding
how a city moves, breathes and evolves.

[EXPLORE CITY →]

● SYSTEM ONLINE

BUILT BY
TANVI × SHARANYA × YASH

NESCOE HACKATHON 2026

■ CITY PULSE

Create a major dashboard section called CITY PULSE. Show the current condition of the city through interactive metric cards for Traffic, Weather, Energy, Green Space, Transport and Waste.

Example demonstration values may include Traffic Flow 72%, Weather 29°C, Energy 438 MW, Urban Green Cover 38%, Network Availability 84%, and Collection Efficiency 76%. These must come from the centralized frontend data source.

■■ CITY NETWORK

Create an interactive-looking city map showing roads, transit routes, important locations, traffic points, green areas, energy nodes and waste collection points. Use CSS/SVG/HTML if a real map library is unnecessary. Add hover/click interactions and animated glowing nodes. Do not add a backend for the map.

■ MOBILITY

Show metro status, bus network, bike sharing, active routes, passenger flow and network availability. Use attractive charts and frontend demonstration data.

■ ENERGY

Show current consumption, renewable contribution, peak demand, daily consumption and energy trend. Create an attractive animated interactive graph.

■ ENVIRONMENT

Show Air Quality Index, Green Cover, Temperature and Humidity with a clear environmental status indicator.

■■ WASTE

Show collection efficiency, recycling rate, active collection vehicles, waste processed and recovery rate using charts or progress indicators.

■■ WEATHER

Show temperature, humidity, wind, weather condition and hourly trend using frontend demonstration data. Do not require an external weather API or API key.

■■ SYSTEM STATUS

TRAFFIC	ONLINE
MOBILITY	ONLINE
ENERGY	ONLINE
ENVIRONMENT	ONLINE
WASTE	ONLINE
NETWORK	ONLINE

Add an overall city health/system score generated from the frontend data.

■ Data Visualization

Use charts only where they improve the interface: line charts, area charts, circular progress, bar charts and network visualizations. Include smooth animations, tooltips, hover states and responsive sizing. Do not turn every empty space into a chart.

■ Interactions

Add smooth scrolling, scroll reveal animations, animated numbers, graph animations, map node animations, navigation highlighting, card hover states, button micro-interactions and subtle background grid movement. Keep animations professional and restrained.

■ Dark / Light Mode

Include a simple dark/light mode toggle. Default to dark. Light mode should retain the NEXUS futuristic identity rather than becoming a completely different website.

■ Responsive Design

The website must work correctly on desktop, laptop, tablet and mobile. Avoid horizontal overflow. Charts must resize correctly and navigation must become mobile-friendly.

■ Suggested React Structure

```
src/
  components/
    Header.jsx
    Hero.jsx
    CityPulse.jsx
    MetricCard.jsx
    CityNetwork.jsx
    Mobility.jsx
    Energy.jsx
    Environment.jsx
    Waste.jsx
    Weather.jsx
    SystemStatus.jsx
    Footer.jsx
  data/
    cityData.js
  App.jsx
  main.jsx
  styles/
```

■ Data Philosophy

Use centralized frontend mock/demo data. This is acceptable because NEXUS is a prototype. Where appropriate, label information as **LIVE DEMO** or **SIMULATED CITY DATA**. Do not pretend the prototype is connected to real municipal infrastructure.

■ Hackathon Presentation Goal

A judge should immediately understand what NEXUS is, what problem it addresses, what city systems it connects, how the dashboard works, how information is visualized, and why a unified interface is useful.

■ Final Quality Checklist

Remove lorem ipsum, unnecessary demo buttons, authentication and backend dependencies. Remove unused packages and console errors. Make navigation work, sections accessible, charts functional, dark/light mode operational, animations stable and the site fully responsive. Ensure the project builds and runs successfully.

FINAL BRANDING

NEXUS

THE CITY, CONNECTED.

BUILT BY
TANVI × SHARANYA × YASH

Final instruction: Build a polished, interactive, futuristic Smart City frontend prototype. Keep the implementation practical, lightweight, frontend-only and hackathon-friendly. Do not turn NEXUS into a massive full-stack application.